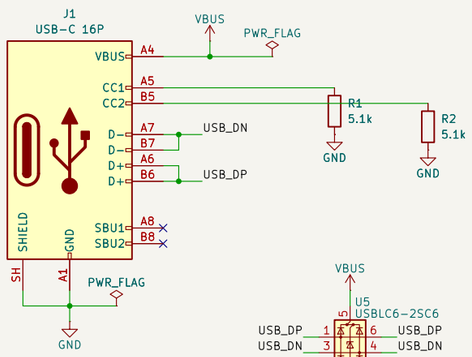
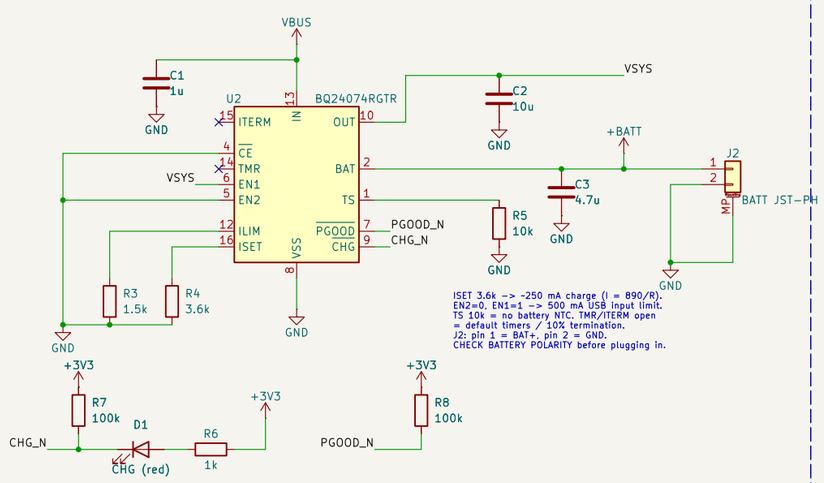
USB-C INPUT (power + native USB data)



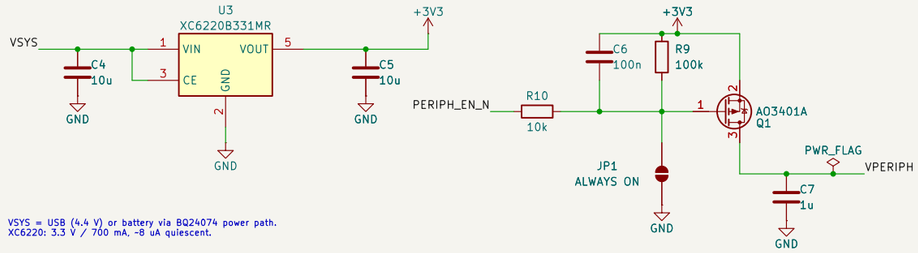
CC1/CC2 5.1k pull-downs: sink, default USB power (500 mA).
D+/D- go straight to the ESP32-C6 USB Serial/JTAG (GPIO13/GPIO12).

LI-ION CHARGER + POWER PATH (BQ24074)



ISET 3.6k -> ~250 mA charge (I = 890/R).
EN2=0, EN1=1 -> 500 mA USB input limit.
TS 10k = no battery NTC. TMR/ITERM open = default timers / 10% termination.
J2: pin 1 = BAT+, pin 2 = GND.
CHECK BATTERY POLARITY before plugging in.

3V3 REGULATOR + SWITCHED PERIPHERAL RAIL

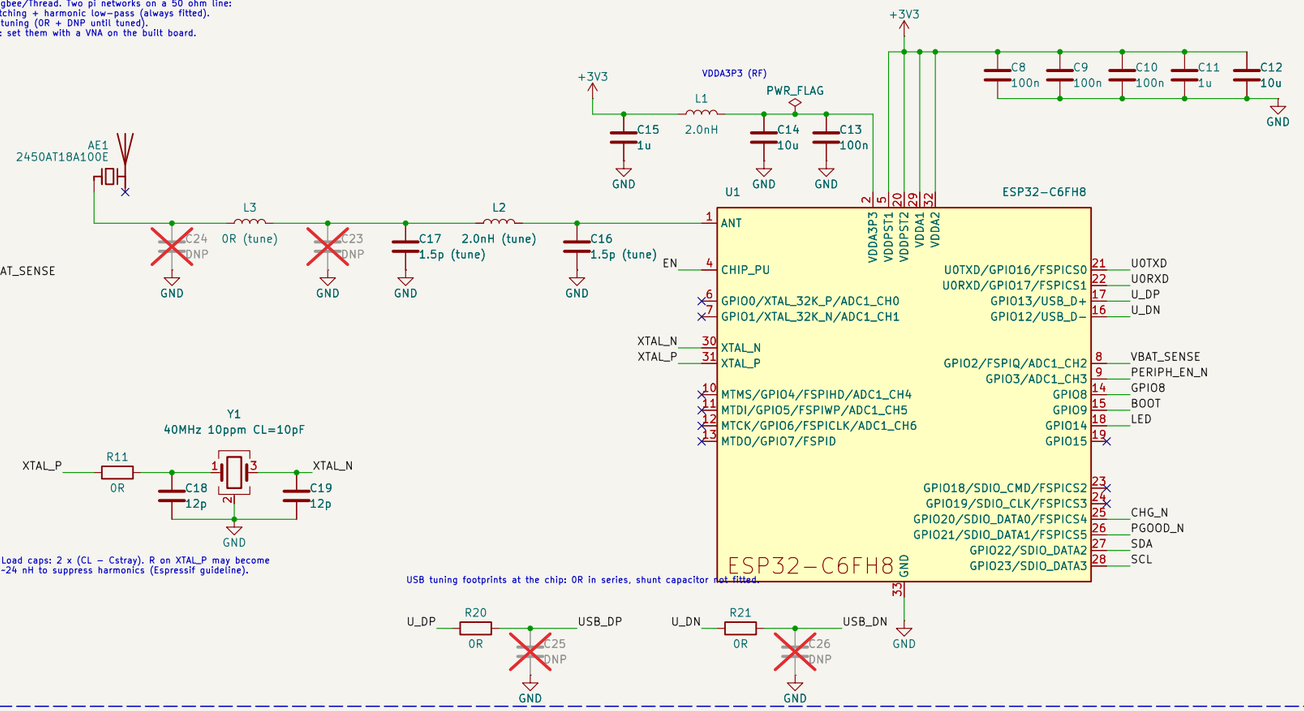


VSYS = USB (4.4 V) or battery via BQ24074 power path.
XC6220: 3.3 V / 700 mA, ~8 uA quiescent.

Q1 switches 3V3 to SHT41 + Qwiic + OLED (VPERIPH).
PERIPH_EN_N (GPIO3) low = on. Off by default and in deep sleep (gate pulled up). Bridge JP1 = always on.
10k/100n gives a soft turn-on (no 3V3 dip).

ESP32-C6FH8 (QFN32, 8 MB in-package flash) — Wi-Fi 6 / BLE / 802.15.4 (Zigbee, Thread)

2.4 GHz chip antenna shared by Wi-Fi and Zigbee/Thread. Two pi networks on a 50 ohm line:
C16-L2-C17 next to the chip pin = chip matching + harmonic low-pass (always fitted).
C23-L3-C24 next to the antenna = antenna tuning (OR + DNP until tuned).
Values marked (tune) are starting points only: set them with a VNA on the built board.

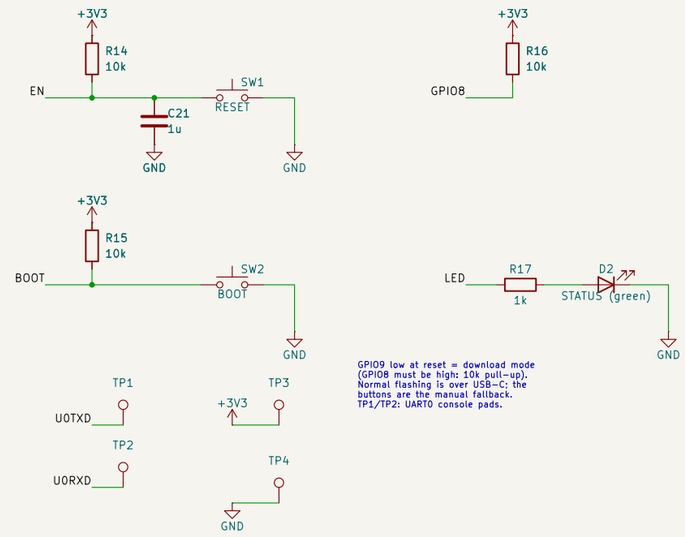


VBAT / 2 to ADC1_CH2 (GPIO2).
~2 uA divider drain.

Load caps: 2 x (CL − Cstray). R on XTAL_P may become ~24 nH to suppress harmonics (Espressif guideline).

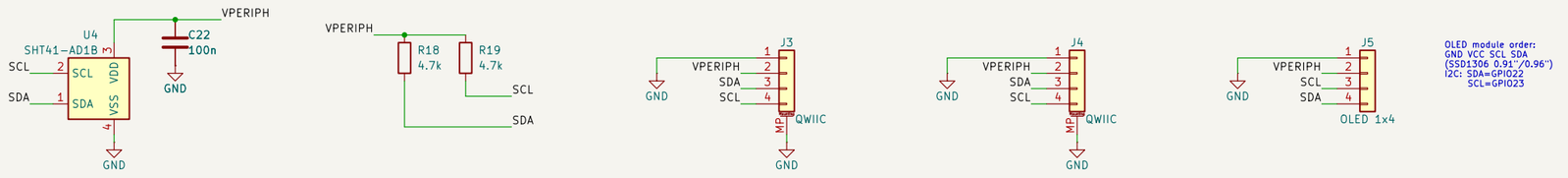
USB tuning footprints at the chip: 0R in series, shunt capacitor not fitted.

RESET / BOOT / STRAPS / STATUS



GPIO9 low at reset = download mode (GPIO8 must be high: 10k pull-up).
Normal flashing is over USB-C; the buttons are the manual fallback.
TP1/TP2: UART0 console pads.

SHT41 + I2C PORTS (all on VPERIPH)



OLED module order:
GND VCC SCL SDA
(SSD1306 0.91"/0.96")
I2C: SDA=GPIO22
SCL=GPIO23

SHT41 layout rules: see README.

Qwiic / STEMMA QT: 1 GND, 2 3V3, 3 SDA, 4 SCL

USB-C charging with power path, 1S Li-ion on JST-PH
ESPHome target: Wi-Fi first, 802.15.4 (Zigbee) capable

Sheet: /
File: esp32c6-sht4x.kicad_sch

Title: ESP32-C6 + SHT41 battery sensor node

Size: A3	Date: 2026-11-10	Rev: B
KiCad E.D.A. 10.0.8		Id: 1/1